

QUERY PROCESSING LAB EXP 1 to 10

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COURSE CODE : DSA0503 - Query Processing

EXP 1

Aim

To write a Python program using Pandas to select and display the distinct Department IDs from the given employee/department data.

Algorithm

- ❖ Start the program.
- ❖ Import the **Pandas** library.
- ❖ Create a DataFrame with the given Department IDs.
- ❖ Select the **DEPARTMENT_ID** column.
- ❖ Use the *drop_duplicates()* or *unique()* method to get distinct Department IDs.
- ❖ Display the distinct Department IDs.
- ❖ Stop the program.

Code

```
import pandas as pd

# Create the DataFrame
data = {
    "DEPARTMENT_ID": [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110,
                      120, 130, 140, 150, 160, 170, 180, 190, 200,
                      210, 220, 230, 240, 250, 260, 270],
    "DEPARTMENT_NAME": [
        "Administration", "Marketing", "Purchasing", "Human Resources",
        "Shipping", "IT", "Public Relations", "Sales", "Executive",
        "Finance", "Accounting", "Treasury", "Corporate Tax",
        "Control And Credit", "Shareholder Services", "Benefits",
        "Manufacturing", "Construction", "Contracting", "Operations",
        "IT Support", "NOC", "IT Helpdesk", "Government Sales",
        "Retail Sales", "Recruiting", "Payroll"
    ],
    "MANAGER_ID": [200, 201, 114, 203, 121, 103, 204, 145, 100, 108,
                   205, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
    "LOCATION_ID": [1700, 1800, 1700, 2400, 1500, 1400, 2700, 2500,
                  1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700,
                  1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700,
                  1700, 1700, 1700]
}

df = pd.DataFrame(data)

# Select distinct Department IDs
distinct_ids = df["DEPARTMENT_ID"].drop_duplicates()

# Display the result
print("Distinct Department IDs:")
print(distinct_ids)
```

OUTPUT

```
IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processing/exp 1 qp.py
Distinct Department IDs:
0      10
1      20
2      30
3      40
4      50
5      60
6      70
7      80
8      90
9     100
10     110
11     120
12     130
13     140
14     150
15     160
16     170
17     180
18     190
19     200
20     210
21     220
22     230
23     240
24     250
25     260
26     270
Name: DEPARTMENT_ID, dtype: int64
>>>
```

EXP 2

Aim

To write a Python program using Pandas to display the Employee IDs of employees who have done two or more jobs in the past.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas library.
- ❖ Create a DataFrame with the employee job history.
- ❖ Group the data by EMPLOYEE_ID.
- ❖ Count the number of jobs for each employee.
- ❖ Select employees whose job count is 2 or more.
- ❖ Display the Employee IDs.
- ❖ Stop the program.

CODE

```
import pandas as pd

# Employee data

df = pd.DataFrame({

    "EMPLOYEE_ID": [102, 101, 101, 201, 114, 122, 200, 176, 176, 200]

})

# Display employees with two or more jobs

result = df["EMPLOYEE_ID"].value_counts()
```

```
print("Employees who have done two or more jobs:")
```

```
print(result[result >= 2])
```

OUTPUT

```
>>> = RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processi
ng/exp 1 qp.py
Employees who have done two or more jobs:
EMPLOYEE_ID
101      2
200      2
176      2
Name: count, dtype: int64
>>> |
```

EXP 3

Aim

To write a Python program using Pandas to display the job details in descending order of Job Title.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas library.
- ❖ Create a DataFrame with the job details.
- ❖ Sort the DataFrame by the JOB_TITLE column in descending order.
- ❖ Display the sorted job details.
- ❖ Stop the program.

CODE

```
import pandas as pd
# Job data
df = pd.DataFrame({
    "JOB_ID":    ["AD_PRE",    "AD_VP",    "AD_ASST",    "FI_MGR",
    "FI_ACCOUNT"],
    "JOB_TITLE": ["President",
                  "Administration Vice President",
                  "Administration Assistant",
                  "Finance Manager",
                  "Accountant"]
})
# Sort by JOB_TITLE in descending order
result = df.sort_values(by="JOB_TITLE", ascending=False)
print(result)
```

OUTPUT

```
>>> |
= RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processi
ng/exp 1 qp.py
      JOB_ID      JOB_TITLE
0  AD_PRES      President
3  FI_MGR      Finance Manager
1  AD_VP  Administration Vice President
2  AD_ASST  Administration Assistant
4  FI_ACCOUNT      Accountant
>>> |
```

EXP 4

Aim

To write a Python program using **Pandas** to create a **line plot** of the historical stock prices of **Alphabet Inc.** between two specific dates.

Algorithm

1. Start the program.
2. Import the **Pandas** and **Matplotlib** libraries.
3. Create a DataFrame with sample stock price data.
4. Convert the **Date** column to datetime format.
5. Plot the **Close Price** against the **Date** using a line graph.
6. Add the title and axis labels.
7. Display the graph.
8. Stop the program.

CODE

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Sample stock price data
```

```
data = {
```

```
    "Date": ["2024-01-01", "2024-01-02", "2024-01-03", "2024-01-04", "2024-01-05"],
```



```
"Close": [140, 142, 141, 145, 147]
}

df = pd.DataFrame(data)

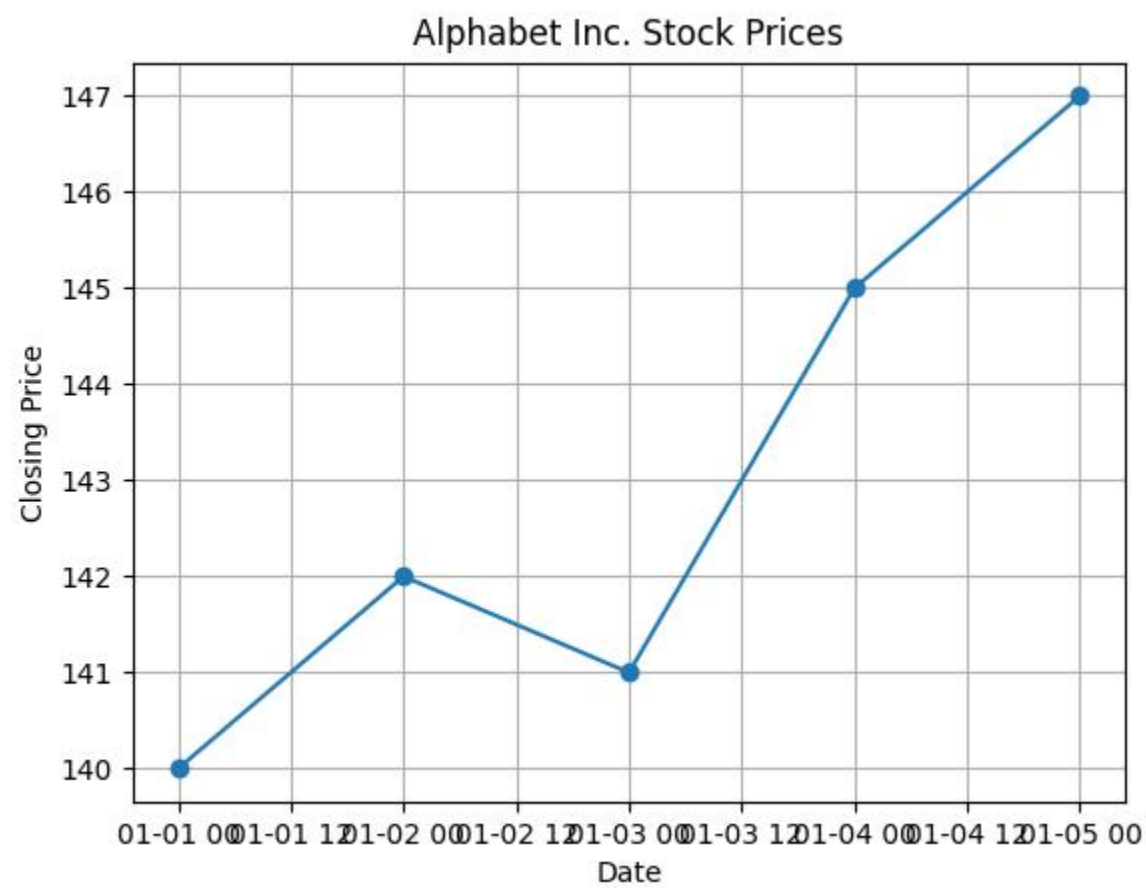
# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])

# Create line plot
plt.plot(df["Date"], df["Close"], marker='o')

plt.title("Alphabet Inc. Stock Prices")
plt.xlabel("Date")
plt.ylabel("Closing Price")
plt.grid(True)

plt.show()
```

OUTPUT



EXP 5

Aim

To write a Python program using Pandas to create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas and Matplotlib libraries.
- ❖ Create a DataFrame with the Date and Trading Volume.
- ❖ Convert the Date column to datetime format.
- ❖ Create a bar plot using the Date and Volume columns.
- ❖ Add the title and axis labels.
- ❖ Display the graph.
- ❖ Stop the program.

CODE

```
import pandas as pd

import matplotlib.pyplot as plt

# Sample stock trading volume data

data = {

    "Date": ["2024-01-01", "2024-01-02", "2024-01-03", "2024-01-04", "2024-01-05"],

    "Volume": [1200000, 1500000, 1000000, 1800000, 1600000]

}
```

```
df = pd.DataFrame(data)

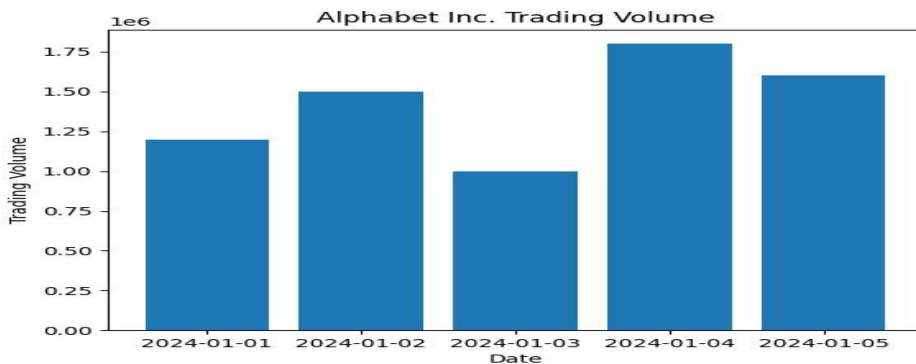
# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])

# Create bar plot
plt.bar(df["Date"].dt.strftime("%Y-%m-%d"), df["Volume"])

plt.title("Alphabet Inc. Trading Volume")
plt.xlabel("Date")
plt.ylabel("Trading Volume")

plt.show()
```

OUTPUT



EXP 6

Aim

To write a Python program using Pandas to create a scatter plot of the trading Volume and Closing Stock Price of Alphabet Inc. stock between two specific dates.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas and Matplotlib libraries.
- ❖ Create a DataFrame with Date, Close, and Volume.
- ❖ Create a scatter plot using Volume on the X-axis and Close Price on the Y-axis.
- ❖ Add the title and axis labels.
- ❖ Display the scatter plot.
- ❖ Stop the program.

CODE

```
import pandas as pd

import matplotlib.pyplot as plt

# Sample stock data

data = {

    "Date": ["01-04-2020", "02-04-2020", "03-04-2020", "06-04-2020", "07-04-2020"],

    "Close": [1105.62, 1120.84, 1097.86, 1186.92, 1186.51],
```

```
"Volume": [1234100, 1964900, 931500, 2064700, 2387300]  
}
```

```
df = pd.DataFrame(data)
```

```
# Scatter Plot
```

```
plt.scatter(df["Volume"], df["Close"])
```

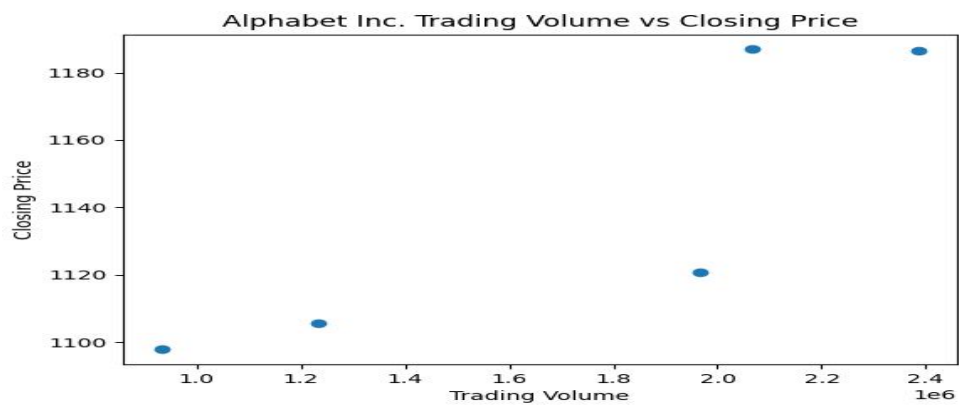
```
plt.title("Alphabet Inc. Trading Volume vs Closing Price")
```

```
plt.xlabel("Trading Volume")
```

```
plt.ylabel("Closing Price")
```

```
plt.show()
```

OUTPUT



EXP 7

Aim

To write a Python program using Pandas to create a Pivot Table and find the maximum and minimum sale values of the items.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas library.
- ❖ Create a DataFrame with Item and Sale columns.
- ❖ Create a Pivot Table using *pivot_table()*.
- ❖ Apply max and min aggregation functions to the Sale column.
- ❖ Display the Pivot Table.
- ❖ Stop the program.

CODE

```
import pandas as pd

# Sample sales data

data = {

    "Item": ["Pen", "Pen", "Book", "Book", "Pencil", "Pencil"],

    "Sale": [100, 150, 300, 250, 80, 120]

}

df = pd.DataFrame(data)
```

```
# Create Pivot Table

pivot = pd.pivot_table(

    df,

    values="Sale",

    index="Item",

    aggfunc=["max", "min"]

)

print(pivot)
```

OUTPUT

```
= RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processi
ng/exp 1 qp.py
      max  min
Sale Sale
Item
Book    300  250
Pen     150  100
Pencil  120   80
```


EXP 8

Aim

To write a Python program using Pandas to create a Pivot Table and find the item-wise units sold.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas library.
- ❖ Create a DataFrame with Item and Units Sold.
- ❖ Create a Pivot Table using the sum function.
- ❖ Display the item-wise units sold.
- ❖ Stop the program.

CODE

```
import pandas as pd

# Sample sales data
data = {
    "Item": ["Pen", "Pen", "Book", "Book", "Pencil", "Pencil"],
    "Units_Sold": [10, 15, 20, 25, 30, 35]
}

df = pd.DataFrame(data)

# Create Pivot Table
pivot = pd.pivot_table(
    df,
    values="Units_Sold",
```

```
    index="Item",  
    aggfunc="sum"  
)
```

```
print(pivot)
```

OUTPUT

```
>>> = RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processi  
ng/exp 1 qp.py  
      Units_Sold  
Item  
Book           45  
Pen            25  
Pencil         65  
>>>
```

EXP 9

Aim

To write a Python program using Pandas to create a Pivot Table and find the total sale amount region-wise, manager-wise, and salesman-wise.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas library.
- ❖ Create a DataFrame with Region, Manager, SalesMan, and Sale_amt.
- ❖ Create a Pivot Table using `pd.pivot_table()`.
- ❖ Set Region, Manager, and SalesMan as the index.
- ❖ Use Sale_amt as the values and sum as the aggregation function.
- ❖ Display the Pivot Table.
- ❖ Stop the program.

CODE

```
import pandas as pd

# Sample sales data
data = {
    "Region": ["East", "Central", "Central", "West", "East"],
    "Manager": ["Martha", "Hermann", "Timothy", "Douglas", "Martha"],
    "SalesMan": ["Alexander", "Shelli", "David", "Michael", "Diana"],
    "Sale_amt": [113810, 25000, 6075, 38336, 14500]
}

df = pd.DataFrame(data)

# Create Pivot Table
pivot = pd.pivot_table(
    df,
    values="Sale_amt",
    index=["Region", "Manager", "SalesMan"],
    aggfunc="sum"
)

print(pivot)
```

OUTPUT

```
>>> = RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processi
ng/exp l qp.py

      Region Manager SalesMan  Sale_amt
Central Hermann Shelli      25000
          Timothy David       6075
East      Martha Alexander  113810
          Diana      14500
West      Douglas Michael   38336
```

EXP 10

Aim

To write a Python program using Pandas to create a DataFrame with 10 rows and 4 columns of random values and highlight negative numbers in red and positive numbers in black.

Algorithm

- ❖ Start the program.
- ❖ Import the Pandas and NumPy libraries.
- ❖ Create a DataFrame with random values.
- ❖ Define a function to color negative numbers red and positive numbers black.
- ❖ Apply the function using `style.map()`.
- ❖ Display the styled DataFrame.
- ❖ Stop the program.

CODE

```
import pandas as pd
import numpy as np

# Create DataFrame with random values
df = pd.DataFrame(
    np.random.randn(10, 4),
```

```

        columns=['A', 'B', 'C', 'D']
    )
# Function to highlight values
def color(x):
    if x < 0:
        return 'color:red'
    else:
        return 'color:black'

# Display styled DataFrame
print(df)

# For Jupyter Notebook
df.style.map(color)

```

OUTPUT

```

>>>
= RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/query processi
ng/exp 1 qp.py
      A      B      C      D
0  0.469949 -0.157052 -0.156593 -0.638053
1  0.978953 -0.716054  0.614020  1.070386
2  0.965893  0.281230 -0.668425 -0.031761
3 -0.576278 -1.123108  1.506916  1.025364
4 -0.239889 -0.650365  0.375837 -0.946248
5 -0.218735 -1.379016  0.147394 -0.733952
6 -0.029388  0.516021  0.157386  1.395421
7 -0.060722  0.758564  0.169183 -0.247198
8  0.917315  0.401463 -0.200316  1.217789
9 -0.858231 -1.264342  1.443099  1.225444
>>>

```